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Gender, Parental Education, and Student Performance: A Statistical Analysis

Application Report

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Repository: <https://github.com/maheat/student-scores-analysis>

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1 Introduction

Understanding which factors are associated with student academic performance is a central concern in educational research and policy. Two factors that have received sustained empirical attention are gender and the socioeconomic background of the family, often proxied by parental level of education (Sirin, 2005). Identifying whether these characteristics are linked to test outcomes can guide targeted interventions and resource allocation in schools.

This report analyses a dataset recording the Mathematics and Language scores of 486 students, provided by the Department of Statistics at TU Dortmund University (Dataset, 2026). Two research questions guide the analysis:

RQ1. Is there a statistically significant difference between male and female students in their Mathematics scores and in their Language scores?

RQ2. Is there a statistically significant difference in scores across the four parental education levels? If so, which pairs of groups differ?

The main findings can be stated at the outset. Gender is significantly associated with scores in both subjects but in opposite directions: male students score higher in Mathematics, while female students score higher in Language. Parental education is also significantly associated with scores; students whose parents hold only a high school diploma consistently score below those from households with any post-secondary qualification.

The remainder of this report is structured as follows. Section 2 describes the dataset. Section 3 defines the statistical methods. Section 4 presents the descriptive and inferential results. Section 5 summarises the findings and discusses limitations.

2 Problem and Data Description

The dataset was provided by the Department of Statistics at TU Dortmund University as part of the application for the Master’s programme in Data Science (Dataset, 2026). It records results for 486 students each assessed in two subjects, yielding 972 observations in long format (one row per student–subject pair). Since each student contributes one Mathematics and one Language score, inferential tests are conducted separately within each subject so that independence holds within each analysis. There are no missing values.

The dataset contains five variables: a unique student identifier, gender (female, male), parental level of education, subject (mathematics, language), and test score on a continuous scale from 0 to 100. The parental education variable is ordinal with four ordered levels: *high school* < *associate’s degree* < *bachelor’s degree* < *master’s degree*.

Of the 486 students, 245 are female and 241 are male. The distribution across parental education levels is: 156 high school, 177 associate’s degree, 94 bachelor’s degree, and 59 master’s degree. Scores range from 19.5 to 100.0.

3 Methods

3.1 Descriptive Statistics

Let $x_1, \dots, x_n$ be a sample of n observed scores within a given group. The **sample mean** and **sample standard deviation** are

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i, \quad s = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2}, \quad (1)$$

measuring average performance and spread around the mean (Montgomery & Runger, 2014). The **median** $\tilde{x}$ is the 0.5-quantile of the ordered sample, and the **interquartile range** is $IQR = Q_3 - Q_1$, where Q_1 and Q_3 are the 25th and 75th sample percentiles.

Box-and-whisker plots display the five-number summary ($\min, Q_1, \tilde{x}, Q_3, \max$) compactly. The box spans $[Q_1, Q_3]$, the line marks the median, and the whiskers extend to the most extreme observations within 1.5IQR of the box edges; points beyond the whiskers are plotted individually as potential outliers (Montgomery & Runger, 2014).

3.2 Welch's Two-Sample t -Test

To address RQ1, a two-sample t -test compares the mean scores of female and male students for each subject separately. The hypotheses are $H_0 : \mu_f = \mu_m$ versus $H_1 : \mu_f \neq \mu_m$, where μ_f and μ_m denote the population means. The test statistic is

$$t = \frac{\bar{x}_f - \bar{x}_m}{\sqrt{s_f^2/n_f + s_m^2/n_m}}. \quad (2)$$

Welch's variant is used because it does not assume equal population variances, making it more generally applicable than Student's t -test (Welch, 1947). The approximate degrees of freedom follow the Welch–Satterthwaite formula

$$\nu = \frac{(s_f^2/n_f + s_m^2/n_m)^2}{(s_f^2/n_f)^2/(n_f - 1) + (s_m^2/n_m)^2/(n_m - 1)}. \quad (3)$$

Under H_0 , $t \sim t(\nu)$, and H_0 is rejected at $\alpha = 0.05$ when $|t| > t_{1-\alpha/2, \nu}$. A 95% confidence interval for $\mu_f - \mu_m$ is $(\bar{x}_f - \bar{x}_m) \pm t_{0.975, \nu} \sqrt{s_f^2/n_f + s_m^2/n_m}$. The practical magnitude of the difference is quantified by **Hedges' g** (Hedges, 1981), a bias-corrected standardised mean difference:

$$g = \frac{\bar{x}_f - \bar{x}_m}{s_p} \left(1 - \frac{3}{4(n_f + n_m - 2) - 1} \right), \quad s_p = \sqrt{\frac{(n_f - 1)s_f^2 + (n_m - 1)s_m^2}{n_f + n_m - 2}}. \quad (4)$$

By convention, $|g| \approx 0.2$ is small, $|g| \approx 0.5$ moderate, and $|g| \approx 0.8$ large (Cohen, 1988).

3.3 One-Way ANOVA and Tukey’s HSD

To address RQ2, a one-way analysis of variance (ANOVA) tests whether mean scores differ across the $k = 4$ parental education groups (Montgomery & Runger, 2014). The model is $Y_{ij} = \mu + \alpha_i + \varepsilon_{ij}$, $\varepsilon_{ij} \stackrel{\text{iid}}{\sim} \mathcal{N}(0, \sigma^2)$, where α_i is the effect of group i and $H_0 : \alpha_1 = \dots = \alpha_4 = 0$. The total variability decomposes as $SS_{\text{total}} = SS_{\text{between}} + SS_{\text{within}}$, where

$$SS_{\text{between}} = \sum_{i=1}^k n_i (\bar{y}_{i\cdot} - \bar{y})^2, \quad SS_{\text{within}} = \sum_{i=1}^k \sum_{j=1}^{n_i} (y_{ij} - \bar{y}_{i\cdot})^2. \quad (5)$$

The F -statistic is

$$F = \frac{SS_{\text{between}} / (k - 1)}{SS_{\text{within}} / (N - k)}, \quad (6)$$

which follows $F(k - 1, N - k)$ under H_0 . The proportion of variance explained is $\eta^2 = SS_{\text{between}} / SS_{\text{total}}$.

When H_0 is rejected, **Tukey’s Honest Significant Difference** test identifies differing pairs while controlling the family-wise error rate at $\alpha = 0.05$ (Tukey, 1949). For each pair (i, i') , $H_0 : \mu_i = \mu_{i'}$ is rejected if

$$|\bar{y}_{i\cdot} - \bar{y}_{i'\cdot}| > \frac{q_{\alpha, k, N-k}}{\sqrt{2}} \sqrt{MS_{\text{within}} \left(\frac{1}{n_i} + \frac{1}{n_{i'}} \right)}, \quad (7)$$

where $q_{\alpha, k, N-k}$ is the upper- α critical value of the studentised range distribution with k groups and $N - k$ error degrees of freedom.

3.4 Assumption Checks

Both the t -test and ANOVA assume approximate normality within groups. The **Shapiro–Wilk test** assesses this formally (Shapiro & Wilk, 1965). Its statistic is

$$W = \frac{\left(\sum_{i=1}^n a_i x_{(i)} \right)^2}{\sum_{i=1}^n (x_i - \bar{x})^2}, \quad (8)$$

where $x_{(i)}$ is the i -th order statistic and the coefficients a_i are derived from the expected values and covariance matrix of standard normal order statistics. Values of W close to 1 indicate normality; small p -values signal departures. **Levene’s test** checks equality of variances by computing an F -statistic from absolute deviations around group means and is robust to non-normality (Montgomery & Runger, 2014). Welch’s t -test is applied regardless of Levene’s outcome, as it remains valid under unequal variances.

4 Evaluation

All analyses were carried out in Python 3 using the packages `pandas` (McKinney, 2010), `numpy` (Harris et al., 2020), `scipy` (Virtanen et al., 2020), `matplotlib` (Hunter, 2007), and `statsmodels` (Seabold & Perktold, 2010). Significance level $\alpha = 0.05$ throughout.

4.1 Descriptive Analysis

Table 1 summarises score distributions by gender and subject. For Mathematics, male students achieve a higher mean of 68.95 compared with 63.86 for female students, a difference of 5.09 points. For Language, the pattern reverses: female students attain a mean of 73.00 against 64.93 for male students, a gap of 8.07 points. Standard deviations are similar across all four groups (13.75–15.26), indicating comparable within-group spread. Figure 1 displays the corresponding boxplots.

Table 1: Score summary by gender and subject.

Subject	Gender	<i>n</i>	$\bar{x}$	$\tilde{x}$	<i>s</i>	Q_1	Q_3	IQR
Mathematics	Female	245	63.86	64.0	15.26	54.0	74.0	20.0
	Male	241	68.95	70.0	14.43	58.0	80.0	22.0
Language	Female	245	73.00	73.0	14.01	64.0	82.5	18.5
	Male	241	64.93	65.5	13.75	54.0	75.0	21.0

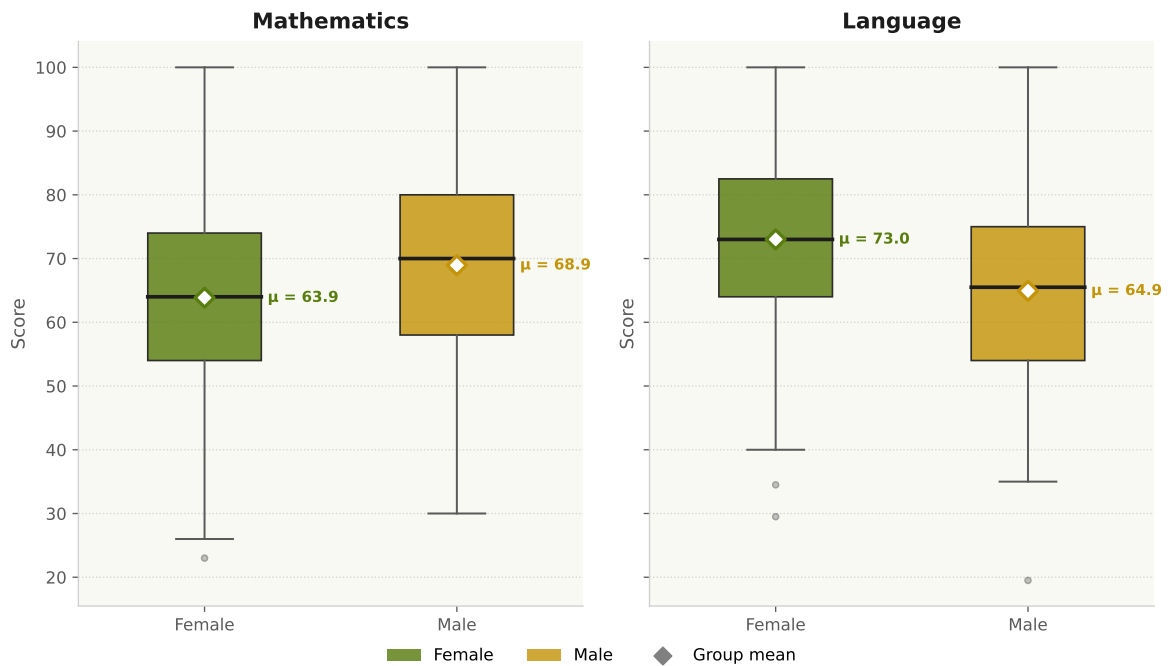


Figure 1: Distribution of test scores by gender, shown separately for Mathematics (left) and Language (right). The central line marks the median; the diamond ($\diamond$) marks the group mean; circles denote outliers.

Table 2 summarises scores by parental education level. For Mathematics, mean scores rise from 62.24 (high school) to 67.81, 68.44 and 69.75 across the three higher levels. For Language, this gradient is steeper: means rise from 63.53 to 70.02, 72.06 and 75.53. In both subjects, the high school group has the lowest mean and the master's degree group the highest. Figure 2 shows the distributions; Figure 3 shows normal Q–Q plots used for assumption checking.

Table 2: Score summary by parental education level and subject.

Subject	Education Level	n	$\bar{x}$	$\tilde{x}$	s	Q_1	Q_3
Mathematics	High school	156	62.24	63.0	14.04	53.8	71.0
	Associate's degree	177	67.81	66.0	15.24	57.0	81.0
	Bachelor's degree	94	68.44	67.0	15.05	61.0	77.8
	Master's degree	59	69.75	73.0	15.15	55.5	81.0
Language	High school	156	63.53	65.0	13.40	53.4	73.0
	Associate's degree	177	70.02	72.0	14.21	59.0	79.0
	Bachelor's degree	94	72.06	73.0	14.27	63.1	80.4
	Master's degree	59	75.53	76.0	13.55	66.3	85.8

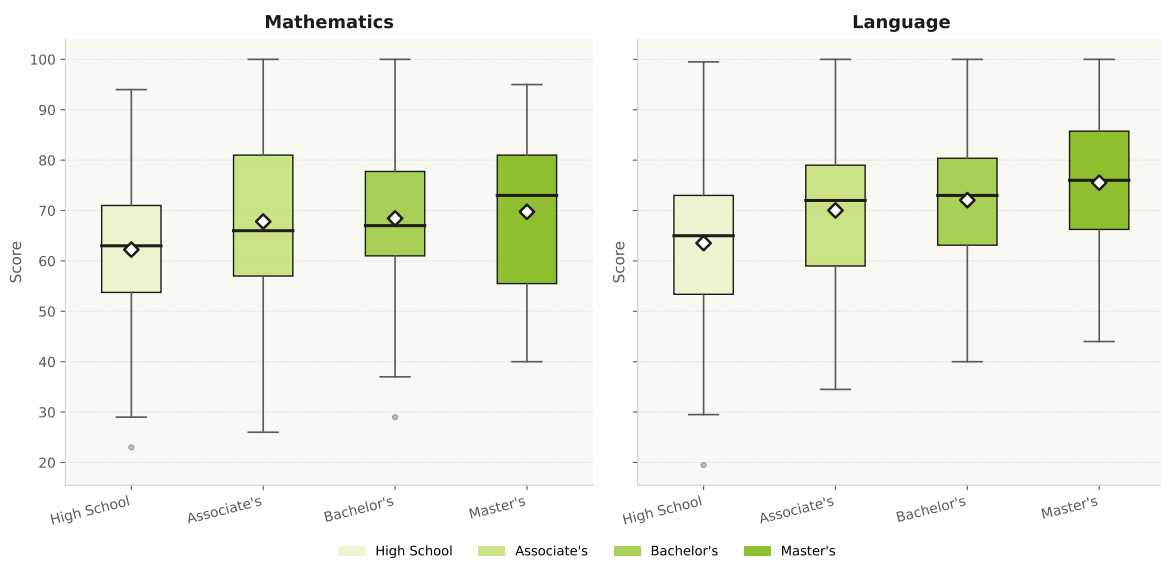


Figure 2: Distribution of test scores by parental education level, shown separately for Mathematics (left) and Language (right). Education levels are ordered from lowest (High School) to highest (Master's Degree). Diamonds ($\diamond$) mark group means.

4.2 Hypothesis Tests

Assumption checks. The Shapiro–Wilk test did not reject normality in any gender group at $\alpha = 0.05$ (all $p > 0.050$). For parental education, mild departures were detected in two Mathematics subgroups (associate's degree: $p = 0.049$; master's degree: $p = 0.032$); all Language subgroups passed. However, with group sizes ranging from 59 to 245, ANOVA is known to be robust to mild departures from normality at these sample sizes (Montgomery & Runger, 2014). Levene's test found no evidence of unequal variances in either gender (Mathematics $p = 0.938$, Language $p = 0.690$) or education (Mathematics $p = 0.282$, Language $p = 0.917$) comparisons. Figure 3 provides a visual confirmation of the normality assumption for the gender groups.

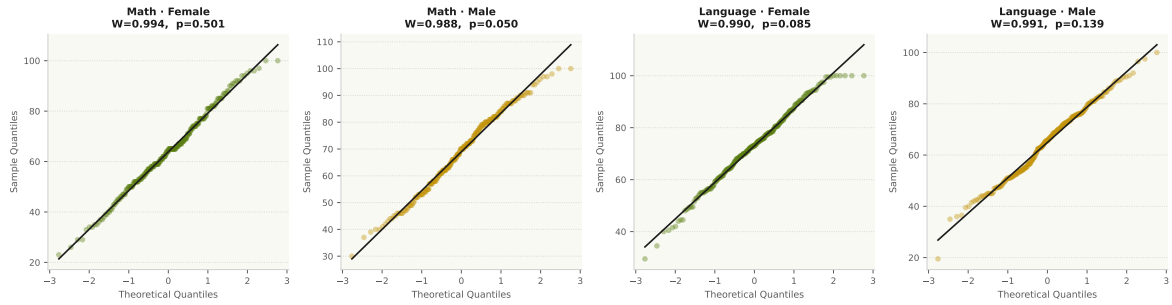


Figure 3: Normal Q-Q plots for all four subject-gender groups. Points on the diagonal indicate exact normality; the Shapiro–Wilk statistic W and p -value are shown above each panel.

RQ1 — Gender differences. Table 3 presents the Welch t -test results. For Mathematics, $t = -3.778$ ($p < 0.001$); the 95% CI $(-7.74, -2.44)$ lies entirely below zero, confirming that male students score higher on average. For Language, $t = 6.410$ ($p < 0.001$); the CI $(5.60, 10.55)$ lies entirely above zero, confirming that female students score higher. Hedges’ $g = -0.342$ for Mathematics indicates a small-to-moderate effect; $g = 0.581$ for Language indicates a moderate effect.

Table 3: Welch t -test results for gender comparison ($\alpha = 0.05$, two-sided). CI_{95} is for $\bar{x}_{\text{female}} - \bar{x}_{\text{male}}$.

Subject	t	p -value	CI_{95}	g	Decision
Mathematics	-3.778	< 0.001	$(-7.74, -2.44)$	-0.342	Reject H_0
Language	6.410	< 0.001	$(5.60, 10.55)$	0.581	Reject H_0

RQ2 — Parental education differences. Table 4 presents the one-way ANOVA results. For Mathematics, $F(3, 482) = 6.223$ ($p < 0.001$, $\eta^2 = 0.037$). For Language, $F(3, 482) = 14.241$ ($p < 0.001$, $\eta^2 = 0.081$). Both ANOVA and the Kruskal–Wallis robustness check (Mathematics: $H = 15.68$, $p = 0.001$; Language: $H = 36.71$, $p < 0.001$) consistently reject H_0 .

Table 4: One-way ANOVA results for parental education comparison ($\alpha = 0.05$).

Subject	$F(3, 482)$	p -value	η^2	Decision
Mathematics	6.223	< 0.001	0.037	Reject H_0
Language	14.241	< 0.001	0.081	Reject H_0

Tables 5 and 6 present the Tukey HSD pairwise comparisons. For **Mathematics**, the high school group scores significantly below all three college groups; no other pairs differ. For **Language**, the high school group again scores significantly below all three college groups, and additionally the master’s degree group scores significantly above the associate’s degree group ($\Delta\bar{x} = 5.50$, $p = 0.043$).

Table 5: Tukey HSD for Mathematics ($\alpha = 0.05$; significant pairs only).

Group 1	Group 2	$\Delta\bar{x}$	p_{adj}	95% CI
Associate's	High school	+5.57	0.004	(1.38, 9.76)
Bachelor's	High school	+6.19	0.008	(1.21, 11.18)
Master's	High school	+7.50	0.006	(1.66, 13.34)

Table 6: Tukey HSD for Language ($\alpha = 0.05$; significant pairs only).

Group 1	Group 2	$\Delta\bar{x}$	p_{adj}	95% CI
Associate's	High school	+6.49	< 0.001	(2.56, 10.42)
Bachelor's	High school	+8.53	< 0.001	(3.85, 13.20)
Master's	High school	+11.99	< 0.001	(6.52, 17.47)
Master's	Associate's	+5.50	0.043	(0.12, 10.89)

5 Summary and Discussion

This report examined whether gender and parental education are associated with student scores in Mathematics and Language.

For **RQ1**, Welch t -tests confirmed significant gender differences in both subjects ($p < 0.001$). Male students scored approximately 5.09 points higher in Mathematics ($g = -0.342$, small-to-moderate effect), while female students scored 8.07 points higher in Language ($g = 0.581$, moderate effect). The direction of the gender gap therefore differs between subjects, consistent with patterns reported in the literature (Hyde et al., 2008).

For **RQ2**, one-way ANOVA detected significant group differences in both Mathematics ($F = 6.22$, $\eta^2 = 0.037$) and Language ($F = 14.24$, $\eta^2 = 0.081$). Tukey HSD tests revealed a consistent threshold effect: students whose parents hold only a high school diploma scored significantly lower than all three college-educated groups in both subjects. The education gradient is stronger in Language, where the master's degree group additionally outperforms the associate's degree group ($p = 0.043$).

Several limitations should be noted. The data are observational, so the findings are associations and cannot be interpreted causally. Unobserved confounders such as household income, school quality, or prior attainment may partly explain the observed patterns. The Shapiro–Wilk test flagged mild departures from normality in two Mathematics subgroups; given the sample sizes and the robustness of ANOVA, these do not affect the reliability of the conclusions (Montgomery & Runger, 2014).

Future work could employ multivariate regression to disentangle the contributions of gender and parental education, or model the two subject scores jointly to account for their within-student correlation.

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